Supplementary material for "Liouville-Preserving Hamiltonian Scattering on Finite Compact Metric Graphs"

The Lean 4 formalization

Philip Hierhager

This supplement describes the Lean 4 development that accompanies the article. Theorem, section and equation numbers without the prefix S refer to the article. Every numbered result of the article is formalized, as are the mathematical claims of Remarks VIII.6, X.5 and X.6 and the worked examples of Sec. XI. Sec. S2 explains what is checked and pairs each result with the declarations that state it; Sec. S3 records how the formal statements encode the printed ones; Sec. S4 lists the proofs that take a different route from the article's; and Sec. S5 records the two corrections to the article that the formalization produced.

S1 The development

The development is a Lean 4 project built against Mathlib, archived at Ref. [1] together with the toolchain pin and the dependency manifest needed to reproduce the build. All statements live in the namespace `HamiltonianMetricGraphs`. A module name `M` below refers to the file `M.lean` in the `HamiltonianMetricGraphs/` directory.

S2 What is checked

What is checked is threefold. *No gaps*: the project compiles with no error and no warning of the `declaration uses 'sorry'` kind, and a search of the sources for `sorry` and `admit` returns no proof term. *No added axioms*: `#print axioms` applied to each main theorem returns exactly {`propext`, `Classical.choice`, `Quot.sound`}, the three axioms of Lean's logic; the file `scripts/Axioms.lean` runs this check. *Statements, not just proofs*: Tables S1 and S2 pair each result with the declarations that state and prove it. Each formal statement can therefore be compared with the printed one.

Table S1: Results of Secs. II–VII, including the definitions and energy lemma added in the revision.

result	declarations	modules
Def. II.1	`FiniteMetricGraph`, `Point`, `graphEDist`, `pathMetric_topology_eq`, `Point.instCompactSpace`	`Basic`, `PotentialExact`, `GraphMetric`
Def. II.2	`IsEdgewiseContDiff`, `isEdgewiseContDiff_two_iff`	`PotentialExact`
Def. II.3	`BoundaryCovector`, `IncomingBoundary`, `OutgoingBoundary`	`Boundary`
Lem. II.4	`boundaryEnergyFormula_eq_hamiltonian`	`PotentialExact`
Def. III.1	`BorelEnergyPreservingScattering`	`BorelScattering`

result	declarations	modules
Prop. III.2	speed_representation_exact	ScatteringExact
Lem. III.3	borel_criterion	ScatteringExact
Def. III.4, Lem. III.5	IsReversibleAt, isReversibleAt_iff_involutive	ScatteringExact
Def. III.6, Lem. III.7	boundaryFluxIn, boundaryFluxIn_apply, map_scattering_boundaryFlux	BoundaryFlux
Examples III.8, III.9	reflectionLaw_spec, transmissionLaw_spec, cyclicLaw_spec	ScatteringExactExamples
Def. IV.1, Lem. IV.2	ScatteringPhaseSpace, measurableSet_saturation	QuotientPhaseSpace, ScatterMap
Def. IV.3	RegularScatteringPhaseSpace, exceptionalEnergies	QuotientPhaseSpace, PhaseSpace
Def. IV.4	IsVertexImpactClass, vertexImpactSet, vertexImpactSetOn	ScatteredTrajectoryLocal, NoZenoExact, ScatteredTrajectoryOn
Lem. IV.5	quotient_charts	ChartsExact
Def. IV.6	quotientLiouville	LiouvilleMeasure
Lem. IV.7	quotientMomentumReversal	QuotientReversal
Lem. V.1	edge_flow_lemma	EdgeFlowExact
Def. V.2	IsScatteredTrajectoryOn, IsScatteredTrajectory	ScatteredTrajectoryOn, ScatteredTrajectory
Lem. V.3	quotientHamiltonian_isScatteredTrajectoryOn, quotientHamiltonian_isScatteredTrajectory	ScatteredTrajectoryOn, NoZenoExact
Lem. V.4	no_zeno_on, no_zeno	ScatteredTrajectoryOn, NoZenoExact
Lem. V.5	impact_local_finiteness_on, impact_local_finiteness	ScatteredTrajectoryOn, NoZenoExact
Prop. V.6	existsUnique_scatteredTrajectoryOn, existsUnique_scatteredTrajectory	ScatteredTrajectoryOn, ScatteredTrajectory
Lem. VI.1	local_flux_form	MeasureExactFluxForm
Lem. VI.2, VI.3	short_time_borel′, one_impact_measure′	MeasureExactShortTime
Lem. VI.4	measurable_flow, flowMeasurableEquiv	FlowMeasurable
Thm. VI.5	globalDeterministicScatteredEvolution, globalDeterministicScatteredEvolution_measure, globalDeterministicScatteredEvolution_measurePreserving	MainTheorem
Cor. VI.6	quotientMomentumReversal_flow	FlowReversal
Prop. VII.1	energy_surface_measure	EnergySurfaceExact
Example VII.2	reflecting_interval_example, reflecting_interval_elastic	EnergySurfaceExact

Table S2: Results of Secs. VIII–XI, added in the revision.

result	declarations	modules
Def. VIII.1, Lem. VIII.2	ShellRegular, shellRegular_iff	Structure, ShellRegularity
Lem. VIII.3	retrace_lemma	RetraceExact
Def. VIII.4	germStep, germStep_eq_speedPermutation	Structure, Bifurcation
Thm. VIII.5	shellGermPerm, regularShell_periodic, flightTime_eq_crossing_or_turning, regularShell_decomposition, regularShell_measure	Structure, ShellTravelTime, ShellMeasure
Rem. VIII.6	remark_disintegration, orbit_pointSpectrum, shell_ergodic_iff_one_cycle	ShellErgodic
Prop. VIII.7	bifurcation_locallyConstant, mem_bifurcationEnergies_of_frequently_cycleType_ne, bifurcation_locallyConstant_analytic	Bifurcation, AnalyticPeriods

result	declarations	modules
Def. IX.1	`RoutingLaw`	`Realization/Routing`
Lem. IX.2	`graph_itinerary`	`Realization/ThinLimitExactItinerary`
Def. IX.3, IX.4, (E), (J), (D)	`ThinHamiltonianNetwork`, `Prepared`, `SatisfiesEdgeReduction`, `JunctionScatteringMap`, `ChannelUniquenessMap`	`Realization/ThinNetwork`, `Realization/ThinHypotheses`
Prop. IX.5	`forced_involution`	`Realization/ForcedInvolution`
Thm. IX.6	`thin_network_limit`	`Realization/ThinLimitExact`
Lem. IX.7, IX.8	`barrier_profile`, `barrier_reflection`	`Realization/StarBarrier`
Thm. IX.9	`explicit_star_junction`	`Realization/StarLimit`
Thm. IX.10	`explicit_energy_dependent_junction`, `explicit_energy_dependent_junction_deterministic`	`Realization/GateLimit`, `Realization/GateDeterminism`
Prop. IX.11	`universal_reversible_realization_paper`	`Realization/UniversalComponent`
Def. X.1, Prop. X.2	`IsMarkovVertexLaw`, `mem_doublyStochastic_iff_vertexLaw`, `preservesGermCounting_iff_flux`, `boundaryKernel_preserves_iff`	`VertexLaw`, `VertexLawExactFlux`
Lem. X.3	`doublyStochastic_normSq_of_unitary`, `exists_doublyStochastic_not_unistochastic`	`VertexLaw`, `VertexLawExact`
Thm. X.4	`birkhoff_vertexLaws`	`VertexLawExact`
Rem. X.5, X.6	`extremePoints_transportPlans_one`, `extreme_plan_not_mapInduced`, `mem_doublyStochastic_of_detailedBalance`, `exists_doublyStochastic_not_detailedBalance`	`VertexLawExact`
Prop. X.7	`coarse_graining`	`Realization/ChannelKernelExact`
Secs. XI A–XI D	`example_threeStar`, `example_theta`, `example_gate_realized`, `example_plateau`	`WorkedExamplesStar`, `WorkedExamplesThetaLimit`, `WorkedExamplesPlateau`
Prop. XI.1	`ribbon_faces`, `ribbon_cyclicStar_dynamic`, `ribbon_theta_same_dynamic`, `ribbon_theta_reversed_dynamic`	`RibbonGraphFaces`, `RibbonGraphExamplesDynamic`

S3 Formal encodings

The formal statements say what the printed ones say. The following choices of encoding are worth knowing when reading them.

- In Definition IV.4, the predicate `IsVertexImpactClass` uses incoming representatives only: every outgoing representative is $S_v\alpha$ for an incoming α, with the same quotient class.
- The open-interval trajectories of Definition V.2 are encoded by `IsScatteredTrajectoryOn`: a total function on $\mathbb{R}$ whose values outside the open interval J are unused. Local finiteness is required only on compact subsets of J. The interval versions in Table S1 prove Lemmas V.3–V.5 on arbitrary open intervals. The original global predicate `IsScatteredTrajectory` has an auxiliary impact set; the lemma `IsScatteredTrajectoryWith.impactSet_eq` identifies it with the vertex-impact set, and `IsScatteredTrajectory.on_univ` relates the global and interval predicates.
- Where a result concerns the flow, the formal statement assumes that Γ has an edge; without one the phase space is empty.
- The path metric of Definition II.1 is the infimum of the lengths of finite chains of steps along edges, the standard definition for graphs; its topology is proved to be the quotient topology of the glued edges.

- The thin networks of Definition IX.3 carry their data for every real ε; only $\varepsilon \in (0, \varepsilon_0)$ enters the hypotheses.
- The invariant measure ν_E of Proposition VII.1 is the image, in the quotient, of the measure $dq/|p|$ on both momentum sheets of each edge; the orbit of a cycle in Theorem VIII.5 (c) is its set of points in the quotient.
- In Proposition XI.1 the genus is the combinatorial genus $g = (2 - \chi)/2$ of the ribbon graph; the development proves that χ is even and at most 2 for a connected ribbon graph, so that g is a natural number. That it is the genus of the associated closed orientable surface is the classification of surfaces, which is not in MATHLIB.
- In Sec. XI C the example is computed for the laws that Theorem IX.10 realizes on the two windows.

S4 Proofs along a different route

Several proofs in the development take a different route from the article's; the statements are unaffected.

- *Lemma V.1*: MATHLIB contains neither Liouville's theorem nor differentiability of the solution of an ODE in its initial condition. The variational equation is solved on short time steps, its Wronskian is identically 1 because $\operatorname{tr} DX = 0$, and the time-t map is the composition of such steps.
- *Lemmas V.3–V.5*: an interval trajectory first agrees locally, and then throughout its interval, with the independently constructed global flow, by edgewise ODE uniqueness and the one-sided scattering limits. Energy conservation and the uniform impact-separation bound then transfer from that flow. The global construction already has its no-Zeno estimate from the recursive edge-flight bounds, so it does not depend on these interval statements.
- *Lemma VI.4*: instead of a Lusin–Souslin selection, a total Borel model of the event chain in end–speed coordinates, proved equal to the recursively defined chain.
- *Lemma VI.1*: the inverse of the flow box is written down explicitly (travel time and speed) and inverted with the inverse function theorem.
- *Lemma VI.3*: derived from the global invariance of the main theorem, which the development proves independently.
- *Lemma VIII.2*: sufficiency through a function that increases at a uniform rate along the flight, instead of the travel-time integral.
- *Proposition VIII.7*: the real-analyticity of the periods through a lemma on analytic dependence of integrals on a parameter and a substitution that removes the singularity at a turning point.
- *Lemma IX.8*: the time bound through a lower bound on the rate of change of $\sqrt{y_* - y}$ below the turning point.

S5 Corrections produced by the formalization

Two imprecisions were found while formalizing. In the submitted version the boundary energy and the edge Hamiltonian were both written with the letter H, and their compatibility was never stated, so the energy-preservation condition of Definition III.1 read as an application of the edge Hamiltonian to a covector. The notation is now distinct and the identity is Lemma II.4. In the revision, Proposition XI.1 needs Γ to have an edge: for a single vertex without edges the Euler formula has no integer genus. The hypothesis is now stated. No other mathematical consequence of the article changes.

References

[1] Philip Hierhager. A formalization of Liouville-preserving Hamiltonian scattering on finite metric graphs in Lean 4. `https://github.com/phierhager/hamiltonian_metric_graphs_lean`, 2026. Lean 4 / Mathlib development, archived snapshot; DOI `<DOI>`, commit `<COMMIT>`.